A Treatise on Differential Equations of the Ordinary Kind

With Notes on Integration

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I am writing this document as my own personal notes regarding analysis and differential equations. I am attempting to cover nets and compactness rigorously in order to precisely explore Integration and Arzela-Ascoli (and related theorems/statements), which are key to the study of Ordinary Differential Equations.

Another topic I am to treat thoroughly is the topic of total variation and bounded variation, which is recurrent with regards to the Riemann-Stieltjes integral.

Textbooks Used to Develop these Notes:

1. ***Principles of Mathematical Analysis*** - *Walter Rudin*
2. ***Mathematical Analysis*** - *Tom Apostol* (for integration)
3. ***General Topology*** - *John Kelley* (for nets)
4. ***Theory of Ordinary Differential Equations*** - *Earl Coddington, Norman Levinson*
5. ***Measure, Integration, and Real Analysus*** - *Sheldon Axler* (for measure theory)

Contents

0 Nets and Sequences

0.1 First Countability

What has been giving us sequential equivalence in $\mathbb{R}$ is the first countability of $\mathbb{R}$ (which means around any point x, there exists a countable basis). We used this to prove the more general properties implied sequential properties (by using the first countable property to construct/apply subsequences).

All metric spaces are first countable; but there are some topological spaces that are not first countable. Consider the following topology on $\mathbb{R}$:

$$\tau \coloneqq \{\varnothing\} \cup \{\mathbb{R} \setminus F : F \text{ is finite}\}$$

Now take some point $x \in \mathbb{R}$ and assume it has some countable basis $\mathcal{B}_x \coloneqq \{B_n\}_{n\in\mathbb{N}}$, in which case:

$$\forall n \in \mathbb{N} : B_n = \mathbb{R} \setminus F_n$$

Then, the following set should be countably infinite:

$$\bigcup_{n\in\mathbb{N}} (\mathbb{R} \setminus B_n)$$

since each F_n is finite. We note that:

$$\bigcup_{n\in\mathbb{N}} (\mathbb{R} \setminus B_n) = \mathbb{R} \setminus \bigcap_{n\in\mathbb{N}} B_n = \mathbb{R} \setminus \{x\}$$

which is a contradiction, since $\mathbb{R}$ is not countably infinite and removing one point does not change this fact. Notice the intersection of the basis elements is solely x. That is because the topology is T_1, which means we can find some open set containing x that separates x and any other arbitrary point. Then by basis definition, there is some $B_n \subseteq O_x$.

This same T_1 condition idea with basis elements gives equivalence of a limit point and ω-limit point in the next section.

In such spaces; if the sequential condition holds, then the general condition holds. E.g.:

1. Sequentially Compact $\implies$ Limit Point Compact
2. Sequentially Closed $\implies$ Topologically Closed

but the same does not hold for the converse.

0.2 Nets . . . Finally

We begin by introducing the notion of a directed set:

Definition 0.1. *We define an ordering relation $\leq$ for the set D and call it a directed set under the following conditions:*

1. *Reflexive:* $\forall x \in D : x \leq x$
2. *Transitive:* $\forall x, y, z \in D : x \leq y \wedge y \leq z \implies x \leq z$
3. *Directed:* $\forall x, y \in D, \exists z \in D : (x \leq z \wedge y \leq z) \vee (z \leq x \wedge z \leq y)$

Notice that the first two properties align with a partial order. However a partial order also requires anti-symmetry, which means:

$$x \leq y \wedge y \leq x \implies x = y$$

which is not required for a directed set. This will become quite important later on. The third property (which is direction) tells us whether the net is upward directed (the left side condition) or downward directed (the right side), or both! We view an equivalent property, for which WLOG, we assume an upward directed set D:

Lemma 0.1. *Given $(D, \leq)$ an upward directed set, then for all finite subsets S of D:*

$$\forall S \subseteq D, \exists z \in D, \forall x \in S : x \leq z \iff \forall x, y \in D, \exists z \in D : x \leq z \wedge y \leq z$$

In other words, all finite subsets of the directed set admit an upper bound.

Proof. $\implies$: The forward direction is trivial as a two element subset is finite.

$\impliedby$: For the converse we prove via induction. The base case is a finite set of two elements, which we are given is true. Now assume it holds for a finite set of n elements such that:

$$S_n \coloneqq \{x_i : 0 \leqslant i \leqslant n-1\} \wedge \forall x_i \in S_n, \exists z \in D : x_i \leq z$$

Now consider $S_{n+1} \coloneqq S_n \cup \{x_n\}$ where x_n is different from all elements in S_n. Then we get that, by the base case:

$$\exists w \in D : x_n \leq w \wedge z \leq w$$

and finally by transitivity, we obtain that $\forall x_i \in S_{n+1} : x_i \leq w$. $\square$

For the remainder of these notes, we assume a directed set D is upward directed unless stated otherwise (the downward directed set notion will also be useful). Now a net is built on a directed set by the following:

Definition 0.2. *A net is a function $\omega : D \to X$ where D is a directed set.*

We define net convergence as the following. Now, to bring about the similarity to sequences, we shall denote a net as $\{x_\alpha\}_{\alpha \in D}$ where D is the directed set. This is still a function where $x_\alpha = \omega(\alpha)$, but this representation shows D as an indexing set.

Now in full generality of an arbitrary topological space, we say that the net converges or that $x_\alpha \to x$ if:

$$\forall U_x \in \boldsymbol{\tau}_X, \exists \alpha_0 \in D, \forall \alpha \geq \alpha_0 : x_\alpha \in U_x$$

where U_x is an arbitrary open neighborhood of the point $x \in X$ with the topology $\boldsymbol{\tau}_X$. For the majority of these notes, we shall work in metric spaces (X, ϱ) (unless stated otherwise), so we can instead say:

$$\forall \epsilon > 0, \exists \alpha_0 \in D, \forall \alpha \geq \alpha_0 : x_\alpha \in B(c, \epsilon) \iff \varrho(x_\alpha, x) < \epsilon$$

so if our directed set $D = \mathbb{N}$, we obtain a sequence. We can alternatively characterize convergence by the following definition:

Definition 0.3. *The tail of a net at α_0 is defined as:*

$$T_{\alpha_0} \coloneqq \{x_\alpha : \alpha \geq \alpha_0\}$$

Then a net converges to some x if and only if $\forall \epsilon > 0, \exists \alpha_0 \in D$ such that:

$$T_{\alpha_0} \subseteq B(x, \epsilon)$$

We say that a net converges if it is eventually in all open neighborhoods of x.

0.3 Sequences/Cluster Points

We introduce a related concept from the lens of sequences:

Definition 0.4. *A sequence frequently visits a set A if:*

$$\forall n_0 \geqslant 0, \exists n \geqslant n_0 : x_n \in A$$

So a cluster point of a sequence is defined as the following:

Definition 0.5. *x is a cluster point of $\{x_n\}_{n\in\mathbb{N}}$ if $\{x_n\}_{n\in\mathbb{N}}$ frequently visits all open neighborhoods of x.*

The notion of a cluster point is something we will bring about when talking about nets and sequences. Considering metric spaces, we get the definition of a cluster point as:

$$\forall \epsilon > 0, \forall n_0 \geqslant 0, \exists n \geqslant n_0 : x_n \in B(x, \epsilon)$$

which permits $x_n = x$ unlike a **limit point of a set**, though there is an eventual connection. We view the following lemma:

Lemma 0.2. *x is a cluster point of $\{x_n\}_{n\in\mathbb{N}} \in X^{\mathbb{N}}$ if and only if $\exists \{n_k\}_{k\in\mathbb{N}} \in \mathbb{N}^{\mathbb{N}} : x_{n_k} \to x$*

Proof. $\Longrightarrow$: If x is a cluster point, then:

$$\forall \epsilon > 0, \forall n_0 \geqslant 0, \exists n \geqslant n_0 : x_n \in B(x, \epsilon)$$

We DO NOT require Axiom of Choice for this proof, as the definition of the cluster point provides us a method of construction. Define the following:

$$\epsilon_m := \frac{1}{m+1}$$

Then, for $m = 0$, choose $n_0 = 0$, which gives some $n \geqslant 0$ such that $x_n \in B(x, \epsilon_0)$. Call this the first term of the subsequence as n_0 (different from the first n_0).

Then for ϵ_1, we can find some $n \geqslant n_0 + 1$ such that $x_n \in B(x, \epsilon_1)$. Repeat this for all $m \in \mathbb{N}$, so that:

$$\forall k \in \mathbb{N} : n_k < n_{k+1} \wedge \forall \epsilon > 0, \exists k_0 \geqslant 0, \forall k \geqslant k_0 : |x_{n_k} - x| < \epsilon$$

because by the Archimedian property, we are guaranteed to find some $m \in \mathbb{N}$ such that:

$$\frac{1}{m+1} \leqslant \epsilon \implies \exists n_m \geqslant 0, \forall n_k \geqslant n_m : |x_{n_k} - x| < \frac{1}{k+1} \leqslant \frac{1}{m+1} \leqslant \epsilon$$

which now shows convergence by Definition 0.2 and can also be phrased in terms of Definition 0.3.

$\Longleftarrow$: If there exists a convergent subsequence, then:

$$\forall \epsilon > 0, \exists n_0 \geqslant 0, \forall n \geqslant n_0 : x_n \in B(x, \epsilon)$$

This lemma shows us that the limit of a sequence is always a cluster point (not necessarily the converse) because convergence is a stricter condition. We see that:

$$\forall \epsilon > 0, \forall n_0' \geqslant 0, \exists n \geqslant n_0' : x_n \in B(x, \epsilon)$$

because if $n_0' < n_0$, then $n = n_0$ satisfies the condition of a cluset point. If $n_0' \geqslant n_0$, then simply choosing $n = n_0'$ satisifes the condition because by convergence $T_{n_0} \subseteq B(x, \epsilon)$. ∎

Notice how (aside from the construction of the subsequence) did not use any properties specific to that of integers. Another thing is the limit of a convergent sequence is a cluster point, but the converse is not necessarily true. This gives rise to nets as we only require the properties of directed sets.

1 Compactness

In the previous section, we discussed notions of convergence and adherence for sequences. We now discuss these notions for sets. Let us view the following theorem:

1.1 Limit Points

We start off with a topological proof, which will be key to discussing limit and adherent points:

Lemma 1.1. *The closure of a set A is equal to all of its adherent points.*

Proof. $\implies$: We prove by contrapositive. If x is not an adherent point, then:

$$\exists \epsilon > 0 : B(x, \epsilon) \cap A = \varnothing \implies B(x, \epsilon) \subseteq X \setminus A$$

So x is an interior point of $X \setminus A$, which means:

$$x \in \text{int}(X \setminus A) = X \setminus \overline{A}$$

$\impliedby$: We once again prove by contrapositive. If $x \notin \overline{A}$, then:

$$x \in X \setminus \overline{A} \subseteq X \setminus A \implies x \in X \setminus A$$

Since $x \setminus \overline{A}$ is open, then x must be an interior point, so:

$$\exists \epsilon > 0 : B(x, r) \subseteq X \setminus A \implies \exists \epsilon > 0 : B(x, r) \cap A = \varnothing$$

□

The limits of sequences ARE NOT necessarily limit points. Given some subset $E \subseteq \mathbb{R}$, x is a limit point of E if:

$$\forall \epsilon > 0 : B(x, \epsilon) \cap E \neq \varnothing$$

Notice how this only requires the intersection to be nonempty; which means there could possibly be a finite number of points in the intersection (but we will see this is not possible). Let us look at the following lemma:

Lemma 1.2. *x is a limit point of E if and only if*

$$\exists \{x_n\}_{n \in \mathbb{N}} \in (E \setminus \{x\})^{\mathbb{N}} : x_n \to x$$

Proof. $\implies$: If x is a limit point, then using the first countability of metric spaces, consider:

$$\epsilon_n := \frac{1}{n+1}$$

Then by axiom of choice, choose $x_n \in B(x, \epsilon_n) \cap E \setminus \{x\}$, so that:

$$\forall n \in \mathbb{N} : |x_n - x| < \frac{1}{n+1} \implies x_n \to x$$

$\impliedby$: Now for the converse. If there exists such a sequence $\{x_n\}_{n \in \mathbb{N}}$ such that:

$$\forall \epsilon > 0, \exists n_0 \geqslant 0, \forall n \geqslant n_0 : |x_n - x| < \epsilon$$

then we see that

$$\forall \epsilon > 0 : B(x, \epsilon) \cap E \setminus \{x\} \neq \varnothing$$

as $\forall n \geqslant n_0 : x_n \in B(x, \epsilon)$ and by definition of the sequence $x_n \in E \setminus \{x\}$. Thus proving equivalence. □

Now we introduce the following notion:

Definition 1.1. *An ω-limit point of a set E is one for which there exists a countably infinite number of points of E within each punctured ϵ-ball around x.*

It follows that every limit point in $\mathbb{R}$ is an ω-limit point from Lemma 1.2, the sequential definition. This is because:

$$\forall \epsilon > 0, \exists n_0 \geqslant 0, \forall n \geqslant n_0 : x_n \in B(x, \epsilon) \cap E \setminus \{x\}$$

and the natural numbers n_0 and beyond are countably infinite (an injection of $\mathbb{N}$ into this set can easily be shown), thus x is an ω-accumulation point. The weakest sufficient condition is a T_1 space for such equivalence.

Now if the set E consists of some sequence $\{x_n\}_{n\in\mathbb{N}}$, then $E \coloneqq \{x_n : n \in \mathbb{N}\}$, we see that x is a limit point of E if and only if there exists a convergent **subsequence** converging to x. This is not immediately trivial as one of the conditions we require for a subsequence is that:

$$\forall \{n_k\}_{k\in\mathbb{N}} : n_k < n_{k+1s}$$

Observe the definition of sequential closure of some subset E:

Definition 1.2. *$\overline{E}$ is the closure of E if and only if*

$$\forall \{x_n\}_{n\in\mathbb{N}} \in E^{\mathbb{N}} : x_n \to x \implies x \in \overline{E}$$

So by the equivalence stated in Lemma 1.2, $\{x_n\}_{n\in\mathbb{N}}$ belongs to $E^{\mathbb{N}}$ as well (it is not guaranteed that $x \in E$), but by sequential closure x or the limit point must belong in $\overline{E}$.

Now steering our attention back to sequences, the limit of a sequence may not be a limit point. Consider an eventually constant or constant sequence. We see that it converges to whatever value it is eventually constant at; but there is an empty ϵ-punctured ball intersection. So a limit point is a limit of a specific type of sequence, but not vice versa.

This distinction will be quite crucial in the following section.

1.2 Limit Points and Sequences

In the previous section, we introduced the notion of a cluster point of a sequence, for which we proved:

$$x \text{ is a cluster point of } \{x_n\}_{n\in\mathbb{N}} \in X^{\mathbb{N}} \iff \exists \{n_k\}_{k\in\mathbb{N}} \in \mathbb{N}^{\mathbb{N}} : x_{n_k} \to x$$

but nothing prevented x_{n_k} from being perhaps a constant sequence such that all terms equal x itself. Now this section focused on limit points of **sets**. So consider the set:

$$S \coloneqq \{x_n : n \in \mathbb{N}\}$$

which is the set of all possible values that appear in the sequence. Notice that the sequence allows for the repetition of values, while S does not. So if S is finite such that:

$$S = \{x_0, ..., x_m\}$$

we notice that because a metric space is T_1, each of these points can be isolated from eachother. Thus, S has no limit point. Now assume S were countabily infinite, which means the sequence has infinitely many distinct terms. Then:

Lemma 1.3.

$$x \textit{ is a limit point of } S \iff \exists \{n_k\}_{k\in\mathbb{N}} \in \mathbb{N}^{\mathbb{N}} : x_{n_k} \to x \wedge (\forall k \in \mathbb{N} : x_{n_k} \neq x)$$

This aligns with Lemma 1.2 with the **extra condition** that it must be a subsequence, which implies:

$$n_k \to \infty \wedge \forall k \in \mathbb{N} : n_k < n_{k+1}$$

Proof. $\implies$: Define $\epsilon_m \coloneqq 1/(m+1)$. Then:

$$\forall k \in \mathbb{N} : B(x, \epsilon_k) \cap S \setminus \{x\} \neq \varnothing$$

Then we have some $x_n \in S$ that belongs to the ball $B(x, \epsilon_k)$. Define this as the n_k term of the "sub"-sequence. Now, this does not necessarily guarantee that the sequence we constructed $\{n_k\}_{k\in\mathbb{N}}$ is necessarily strictly increasing.

$\impliedby$: If there exists a convergent subsequence, then:

$$\forall \epsilon > 0, \exists k_0 \geqslant 0, \forall k \geqslant k_0 : x_{n_k} \in B(x, \epsilon)$$

Now given all terms of the subsequence do not equal x and S is countably infinite, we see that:

$$\forall \epsilon > 0 : B(x, \epsilon) \cap S \setminus \{x\} \neq \varnothing$$

and furthermore x is an ω-limit point. $\square$

This lemma is the engine behind the equivalence of limit point compactness and sequential compactness (the forward direction utilizes first countability).

Going forward, we shall separate the terminologies so that we talk about cluster points of sequences/nets and limit/adherent points of sets. We provide a lemma for their equivalence below:

Another lemma requiring a stronger sufficient condition is the following:

Lemma 1.4. *If $\{x_n\}_{n\in\mathbb{N}} \in X^{\mathbb{N}}$ consists of all DISTINCT terms (each term appears at most once), then:*

$$x \textit{ is a limit point of } S \iff x \textit{ is a cluster point of } \{x_n\}_{n\in\mathbb{N}}$$

Proof. Lemma 0.2 provides us with the equivalence that x is a cluster point of the sequence if and only if there exists a convergent subsequence to x. Now since the sequence itself has all distinct points, so does the convergent subsequence.

One of the terms could perhaps be x, but since it appears at most once, omission of a finite number of points does not change convergence of the sequence. We prove this simply below. Let x be the n_j-th element of the original subsequence derived from the definition of cluster point. Then, for the new subsequence with x omitted:

$$\forall \epsilon > 0, \exists k_0' \coloneqq \max\{k_0, j+1\}, \forall k \geqslant k_0' : |x_{n_k} - x| < \epsilon$$

where k_0 comes from the convergence of the subsequence containing x. Thus we can always find a tail of the new sequence within the ϵ-neighborhood of x. If $k_0 < j$, then $k_0' = j+1$, and by convergence of the initial subsequence, the new one certainly converges as:

$$T'_{k_0'} = T_{j+1} \subseteq T_{k_0} \subseteq B(x, \epsilon)$$

In the case that $k_0 > j$, we see that:

$$T'_{k_0'} = T_{k_0} \subseteq B(x, \epsilon)$$

$\square$

Lemma 1.4 and 1.5 will be key when discussing sequential compactness.

1.3 Sequential Compactness

Theorem 1.1 (Sequential Bolzano Weierstrass)**.** *Any bounded sequence has a convergent subsequence.*

To prove this fact, we show first that any sequence in $\mathbb{R}$ has a monotonic subsequence.

Proof. Let us define the following set, we call "peaks" of a sequence:

$$P \coloneqq \{n \in \mathbb{N} : (\forall j > n : x_j < x_n)\}$$

In the case that P is countably infinite, then we set:

$$n_0 \coloneqq \inf P \wedge \forall k > 0 : n_k \coloneqq \inf\left(P \setminus \{n_0, ..., n_{k-1}\}\right)$$

In which case we are guaranteed that $\forall k \in \mathbb{N} : n_{k+1} > n_k$ which tells us that $x_{n_{k+1}} < x_{n_k}$, showing that there exists a strictly decreasing subsequence. In the case that P is finite, we choose the following:

$$n_0 \coloneqq \sup(P) + 1$$

that way the peaks are fully ommited from the subsequence, as they will prove to be problematic in the construction. Then for any $k \in \mathbb{N}$, we are guaranteed that $n_k \notin P$ (since $\{n_k\}_{k\in\mathbb{N}}$ is strictly increasing and $n_0 \notin P$), which means:

$$\exists j > n_k : x_j \geqslant x_{n_k}$$

So we simple define:

$$\forall k \in \mathbb{N} : n_{k+1} \coloneqq \inf\{j > n_k : x_j \geqslant x_{n_k}\}$$

in which case we obtain a non-decreasing subsequence. Now any bounded sequence has a monotone subsequence, which converges. What this means is given some sequence $\{x_n\}_{n\in\mathbb{N}}$, we have that:

$$\exists y \in \mathbb{R}, \exists r > 0, \forall n \in \mathbb{N} : x_n \in B(y, r)$$

so the sequence has a convergent subsequence within the ball, but such a subsequence may not converge within the ball. This is due to the topological idea that the limit point certainly belongs in the closure of the set but not necessarily in the open set. Consider the bounded sequence:

$$\{x_n\}_{n\in\mathbb{N}} \in (0,1)^{\mathbb{N}} : \left(\forall n \in \mathbb{N} : x_n \coloneqq \frac{1}{n+2}\right)$$

Then $x_n \to 0$ yet $0 \notin (0, 1)$. This idea of closedness plays a key role in sequential compactness. $\square$

Definition 1.3. *A sequentially compact interval of $\mathbb{R}$ is one for which every sequence has a convergent subsequence that converges to a point within the interval.*

The previous Theorem 1.1 leads to a key theorem:

Theorem 1.2. *An interval of $\mathbb{R}$ is sequentially compact if and only if it is closed and bounded.*

$\Longleftarrow$: This direction follows from Sequential Bolzano Weierstrass. If the interval is bounded, then any sequence has a convergent monotone subsequence. Since it is closed, the limit of the sequence, which is an adherent point, belongs in the closure.

$\implies$: Consider any convergent sequence $\{x_n\}_{n\in\mathbb{N}}$ such that $\forall n \in \mathbb{N} : x_n \in I$. Then by sequential compactness:

$$\exists \{n_k\}_{k\in\mathbb{N}} : x_{n_k} \to x \wedge x \in I$$

which forces $x_n \to x$. Thus all convergent sequences converge to points within the set (which by sequential closure) means I is closed. Now assume the interval is not bounded, in which case:

$$\forall r > 0, \forall y \in \mathbb{R} : I \nsubseteq B(y, r)$$

Or equivalently (to bring about the sequential format, we use first countability):

$$\forall n \in \mathbb{N}, \forall y \in \mathbb{R}, \exists x \in I : x \notin B(y, n+1)$$

Choose some arbitrary $y \in \mathbb{R}$. Then $\forall n \in \mathbb{N}$ choose x_n such that $x_n \notin B(y, n+1)$, so that:

$$\forall n \in \mathbb{N} : |x_n - y| > n + 1 > 1/2$$

Now assume there existed some z such that some subsequence indexed by $\{n_k\}_{k\in\mathbb{N}}$ converged to z. Then:

$$\forall k \in \mathbb{N} : |x_{n_k} - y| \leqslant |x_{n_k} - z| + |y - z|$$

$$\implies \forall k \in \mathbb{N} : ||x_{n_k} - y| - |y - z|| \leqslant |x_{n_k} - z|$$

Taking limit $k \to \infty$ we see that $n_k \to \infty$, thus:

$$||x_{n_k} - y| - |y - z|| \to 0$$

which is a contradiction because $|x_{n_k} - y| > n_k + 1$ which tells us that $|x_{n_k} - y| \to \infty$, so $x_{n_k} \nrightarrow z$ and since z is arbitrary, this holds true for all such $z \in \mathbb{R}$.

Lemma 1.5. *For any set $I \subseteq \mathbb{R}$, I is bounded if and only if it is totally bounded (ONLY for $\mathbb{R}$).*

Proof. $\impliedby$: If I is bounded, then:

$$\exists r > 0, \exists y \in \mathbb{R} : I \subseteq B(y, r) \iff \exists r > 0, \exists y \in \mathbb{R}, \forall x \in I : x \in B(y, r)$$

We aim to show I is totally bounded, which means:

$$\forall \epsilon > 0, \exists x_0, ..., x_m \in I : I \subseteq \bigcup_{i=0}^{m} B(x_i, \epsilon)$$

This converse direction is necessarily true; however, the forward direction is not necessarily true in all spaces. Consider the following:

$$M := \max\{|x_i - x_j| : 0 \leqslant i, j \leqslant m\}$$

Then, choose some x_i; so without loss of generality, we choose x_0:

$$\forall x \in I, \exists x_i : x \in B(x_i, \epsilon) \implies |x - x_0| \leqslant |x - x_i| + |x_i - x_0| < \epsilon + M$$

$\implies$: Now to prove the forward direction. We have that I is bounded, which means:

$$\exists y \in \mathbb{R}, \exists r > 0 : I \subseteq B(y, r)$$

Now we can show that $B(y, r)$ is totally bounded. Simply consider the uniform partition:

$$\Pi_B := \left\{ (y - r) + \frac{2r}{n} i : 0 \leqslant i \leqslant n \right\}$$

whereby:

$$\exists n \in \mathbb{N} : \frac{2r}{n} < \frac{\epsilon}{2}$$

$$\implies \forall \epsilon > 0, \exists \Pi_B : I \subseteq B(y, r) \subseteq \bigcup_{x_i \in \Pi_B} B(x_i, \epsilon/2)$$

Now $\forall x_i \in \Pi_B$ where $B(x_i, \epsilon/2) \cap I \neq \varnothing$, choose x_i' such that:

$$x_i' \in B(x_i, \epsilon/2) \cap I$$

and add it to the partition Π_I. Then we observe that:

$$\forall x \in I, \exists x_i \in \Pi_B : x \in B(x_i, \epsilon/2)$$

$$\implies \exists x_i' \in \Pi_I : |x - x_i'| \leqslant |x - x_i| + |x_i - x_i'| < \epsilon/2 + \epsilon/2 = \epsilon$$

thus, I is totally bounded such that,

$$\forall \epsilon > 0, \exists \Pi_I : I \subseteq \bigcup_{x_i' \in \Pi_I} B(x_i', \epsilon)$$

$\square$

Now the main assumption we made here is that every open ball in $\mathbb{R}$ is totally bounded (which we can show via uniform partitions). The rest of the argument is not unique to $\mathbb{R}$. This can give insight into what spaces obey this Heine-Borel compactness property.

1.3.1 Bounded/Totally-Bounded Equivalence

NOTE: I will look into this later, it has something to do with finite dimensions/functional analysis. The main thing to investigate is in what spaces are open balls totally bounded.

1.4 Limit Point Compactness

We now discuss an equivalent form of compactness called limit point compactness (which due to the first countability of $\mathbb{R}$) is essentially equivalent to the sequential characterization.

Definition 1.4. *A set $K \subseteq \mathbb{R}$ is limit point compact if for all infinite (countably or uncountably) sets $E \subseteq K$, there exists a limit point of E in K.*

Theorem 1.3. *A set K is limit point compact if and only if it is sequentially compact in $\mathbb{R}$.*

$\implies$: If K is limit point compact, then every infinite subset is limit point compact. Then consider each sequence $\{x_n\}_{n\in\mathbb{N}} \in K^{\mathbb{N}}$ and define:

$$S := \{x_n : n \in \mathbb{N}\}$$

If S is finite, then by pigeonhole principle, we can find a constant convergent subsequence. If S is countably infinite, then there exists some point $x \in K$ which is a limit point of S. By Lemma 1.3, we have the existence of a subsequence with all terms being different from x converging to x.

$\impliedby$: Consider every infinite subset of K. From that we can always find a countably infinite subset. Let us define this subset as E. Then consider an arbitrary sequence $\{x_n\}_{n\in\mathbb{N}} \in E^{\mathbb{N}}$, such that $S := \{x_n : n \in \mathbb{N}\}$ is countably infinite and all terms of the sequence are unique (this is possible because E is countably infinite).

Then, the sequence must admit a convergent subsequence, which means it admits a cluster point. By Lemma 1.4, we see that this cluster point must be a limit point of $S \subseteq E$, thereby a limit point of E.

2 Topological Compactness

Open covers give us a way to talk about topological compactness in a highly geometric fashion. The two previous notions, limit point and sequential compactness are quite difficult to intuitively understand geometrically.

Definition 2.1. *Given some subset E of a topological space X, an open cover of E is a collection:*

$$\{O_\alpha : \alpha \in I \wedge O_\alpha \in \tau_X\}$$

such that:

$$E \subseteq \bigcup_{\alpha \in I} O_\alpha$$

We say that a subset E of a topological space X (which can include X itself) is topologically compact if EVERY open cover has a finite subcover. By the Heine-Borel definition in $\mathbb{R}$, we see that $(0, 1)$ is not compact. For this, define the open cover:

$$\left\{ \left(0, 1 - \frac{1}{n+1}\right) : n \in \mathbb{N} \right\}$$

which admits no finite subcover. Consider any finite subcollection of this open cover. We see that its union must be some interval $(0, 1 - 1/(m + 1))$ for some $m \in \mathbb{N}$ (the largest m). However, we are guaranteed to always find some real number r such that:

$$1 - \frac{1}{m+1} < r < 1$$

which tells us that there cannot be any finite subcover.

Notice that if a metric space is topologically compact, then it must be totally bounded, as consider the following cover for some fixed $\epsilon > 0$ and $E \subseteq X$:

$$\{B(x, \epsilon) : x \in E\}$$

Since X is topologically compact, it admits a finite subcover, which means:

$$\exists x_0, ..., x_m \in E : E \subseteq \bigcup_{0 \leqslant i \leqslant m} B(x_i, \epsilon)$$

The converse is not necessarily true, as evident by our example with $(0, 1)$. Because it is bounded, by Lemma 1.5, it is totally bounded (this is true only for spaces where open balls are totally bounded (possibly?)), yet we showed it is neither Heine-Borel compact, nor topologically compact.

2.1 Topological and Limit Point Compactness

3 Integration and Nets

4 Arzela-Ascoli and Nets